SPIT Cache Module

Phase 4 Engineering Roadmap: Distributed Clustering, Raft Consensus & RESP3 Protocol

Institution: Sardar Patel Institute of Technology (SPIT)	**Academic Year:** 2026–27 | Semester: V
Department: Computer Science & Engineering	**Subject:** Mini Project (PR1) — Phase 4 Roadmap
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Strategic Vision: *While Phase 2 established UI/UX modeling and Phase 3 completed the single-node persistent core (WAL + Snapshots + Two-Phase Recovery),* ***Phase 4 transitions SPIT Cache into an enterprise-grade distributed system*** *capable of horizontal sharding, multi-node Raft consensus replication, and seamless integration with existing Redis client libraries.*

1. Phase 4 Objectives & Strategic Scope

Phase 4 addresses the single-node capacity and single-point-of-failure constraints of Phase 3. The primary engineering goals are:

- **High Availability (HA):** Zero downtime failover when an active master node crashes via automated leader election.
- **Horizontal Scalability:** Partitioning cache keys across an elastic ring of cluster nodes using Consistent Hashing.
- **Consensus Replication:** Strong distributed durability via Raft quorum commits across 3 to 5 storage nodes.
- **Ecosystem Interoperability:** Redis RESP3 wire protocol compatibility to allow drop-in replacement in existing enterprise stacks.

2. Architectural Evolution: Single-Node to Distributed Cluster

Subsystem Dimension	Phase 3 (Single-Node Core)	Phase 4 (Clustered Architecture)
Topology	Standalone single-process engine	3-to-5 Node Raft Replicated Cluster
Durability Scope	Local disk append (wal.log + snapshot.bin)	Distributed Replicated Log (Quorum Commits)
Capacity Limit	Bounded by single host RAM	Distributed RAM partitioning across cluster nodes
Client Interface	HTTP/REST & Server-Sent Events	Redis RESP3 TCP Socket + HTTP Gateway
Failure Model	Crash recovery via local log replay	Automated sub-second leader election failover
Sharding	No sharding (monolithic memory index)	Consistent Hashing Ring with 128 Virtual Nodes

3. Core Technical Pillars of Phase 4 Implementation

Pillar A: Raft Consensus Replicated State Machine

The single-node `WalManager` will be upgraded into a distributed Raft consensus engine. Each node maintains three internal states: *Follower*, *Candidate*, or *Leader*. Heartbeats (every 150ms) monitor node liveness. Write transactions are acknowledged to clients only after a quorum of nodes ($Q = \lfloor N/2 \rfloor + 1$) have successfully appended the record to their local WAL.

Pillar B: Consistent Hashing & Virtual Nodes

To distribute keys evenly without full reshuffling when nodes join or leave, a 32-bit consistent hashing ring will be deployed. Each physical server maps to 128 virtual nodes (tokens) on the ring. Key routing uses MD5 or Murmur3 hashes to find the successor node on the ring in $O(\log V)$ time via binary search.

Pillar C: Redis Serialization Protocol (RESP3) Adapter

A native TCP listener on port 6379 will decode RESP3 frames. Commands supported will include: GET, SET, DEL, EXPIRE, TTL, PING, INFO, and CLUSTER NODES. This enables existing production microservices built in Java, Python, Go, and Node.js to use SPIT Cache as an official Redis alternative without changing client drivers.

Pillar D: Chaos Engineering & Fault Injection Testbed

To satisfy rigorous verification criteria, Phase 4 will introduce automated network partition simulations (split-brain scenarios), abrupt leader node kills, and packet delay injection using Jepsen-style automated test harnesses.

4. Work Breakdown Structure & Milestones (16-Week Plan)

Sprint	Core Deliverable	Technical Focus & Verification	Nature
Sprints 1–3 (W1–6)	Raft Consensus Core	Leader election, AppendEntries RPC, log replication across 3 nodes	Mandatory
Sprints 4–5 (W7–10)	Consistent Hashing Ring	Virtual node distribution, ring topology, key rebalancing migrations	Mandatory
Sprints 6–7 (W11–13)	RESP3 Wire Protocol	TCP socket server on 6379, pipelining, redis-cli compatibility	Mandatory
Sprint 8 (W14–15)	Cluster Telemetry Console	Visual topology map, active leader indicators, node partition graph	Mandatory
Sprint 9 (W16)	Chaos & Performance Suite	Split-brain tests, recovery benchmarks, final project documentation	Mandatory
Extension A	Zero-Copy File I/O	Memory-mapped files (mmap) for WAL flushing	Optional
Extension B	Tiered NVMe Storage	Spillover of cold LRU keys to SSD rather than deletion	Optional

5. Expected Evaluation Outcomes & Deliverables

- **Benchmark Targets:** Sustained throughput of $50,000+\text{ ops/sec}$ across 3 clustered nodes with sub-2ms $p99$ latency.
- **Fault Recovery Guarantee:** Automated leader election completing in under 350ms upon leader node death.
- **Academic Artifacts:** Research paper draft for IEEE/Springer submission on lightweight persistent in-memory caching.